

Bhaskar Mitra

INDEPENDENT RESEARCHER

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Summary

I am an information retrieval researcher based in Tiohtià:ke / Montréal, Canada. My research focuses on AI-mediated online information access and questions of social justice and emancipation in the context of these sociotechnical systems. I have two decades of industry experience and currently working as an independent researcher. I am also currently serving as the ACM SIGIR Secretary and as an Associate Editor for the ACM Transactions on Information System (TOIS) journal. I previously served as the ACM SIGIR Community Relations Coordinator (2022-2025). I co-founded several community benchmarking efforts including the MS MARCO ranking leaderboards, two TREC Tracks co-sponsored by the US National Institute of Standards and Technology (NIST): the Deep Learning Track (2019-2023) and the Tip-of-the-Tongue Track (2023-2025), and the NTCIR Tip-of-the-Tongue Task (2026-). I have received several awards for my research, including two ACM SIGIR Early Career Researcher Awards (2024) for excellence in research and in community engagement.

Education

University College London

London, UK

PHD IN COMPUTER SCIENCE

2016 - 2021

- Thesis: "Neural Methods for Effective, Efficient, and Exposure-Aware Information Retrieval".
- Advisor: Dr. Emine Yilmaz.
- Awarded the Dean's Prize Scholarship for the full course of the program.

Visvesvaraya Technological University

Belgaum, India

BE IN COMPUTER SCIENCE

2002 - 2006

- Graduated with First Class Honors.

Employment Experience

Microsoft Research

Canada

PRINCIPAL RESEARCHER

2022 - 2025

- Conducted machine learning research on enterprise knowledge access with strong focus on both academic and product impact.
- Developed state-of-the-art machine learning methods for enterprise knowledge extraction, entity linking, generative modeling of structured knowledge, and knowledge-base augmentation for large language models in Project Alexandria.
- Critically explored the consequences of AI-mediated enterprise knowledge access systems and their impact on workers and society.
- Actively contributed to company-wide strategic initiatives around AI & Society / Responsible AI.

Microsoft

India, USA, UK, & Canada

PRINCIPAL APPLIED SCIENTIST

2006 - 2022

- Conducted machine learning and information retrieval research at Bing with strong focus on societal impact.
- Shipped several search quality improvements for Bing and contributed to strategic external-facing initiatives like msmarco.org and open source projects such as cmtk.ai.
- Recipient of multiple internal awards incl. the High-Potential employee program (2007 & 2008), the Gold Star Award (2010), the Above and Beyond Awards (2012), a Special Cash Award (2017), a Special Stock Award (2020), and multiple service awards.
- International Work experience across multiple labs: India (2006 - 2010), USA (2010 - 2013), UK (2013 - 2018) and Canada (2018 - 2022).

Amazon

India

INTERN

2006

- Worked on distributed data store and query management components of an early Amazon Payments Web Service prototype.

Hewlett-Packard

India

INTERN

2005

- Worked on HP-UX Kernel Debugger to surface kernel crash dump data using AJAX based hypertext interface.

Honors & Awards

2024	ACM SIGIR Early Career Researcher Award , for excellence in Research	USA
2024	ACM SIGIR Early Career Researcher Award , for excellence in Community Engagement	USA
2021	Microsoft Service Award , for completing 15 years at Microsoft	Canada

2020	Best Full Paper Nominee , CIKM 2020	Online
2018	Best Short Paper Award , SIGIR 2018	USA
2018	Ranked 7th , RecSys Challenge 2018	Canada
2017	Outstanding Reviewer Award , SIGIR 2017	Japan
2016	Dean's Prize scholarship , University College London	UK
2016	Microsoft Service Award , for completing 10 years at Microsoft	UK
2011	Microsoft Service Award , for completing 5 years at Microsoft	USA
2011	Microsoft IDC Innovation Award , Microsoft India Development Center	India
2010	Microsoft IDC Innovation Award , Microsoft India Development Center	India
2006	Semi-finalist , Code4Bill competition, all India national coding competition organized by Microsoft	India
2001	Gold merit certificate , for securing all India rank 6 and state level rank 1 in the 1st National Cyber Olympiad organized by The National Science Olympiad Foundation of India	India
2001	Ranked 1st , in The Academic & Achievement Tests jointly conducted by The Council for the Indian School Certificate Examinations and The Institute of Psychological & Educational Measurement	India
2001	High Distinction and Distinction , in Mathematics and Science in The International Assessments for Schools organized by The University of New South Wales and The Educational Testing Centre	India
1996	High Distinction and Distinction , in Mathematics and Science in The International Assessments for Schools organized by The University of New South Wales and The Educational Testing Centre	India

In news and media

- 2025 Ariella Steinhorn. *Machine learning expert: "AI hype" is the true product, to provide cover for social agendas*. In Hard Reset, published: <https://www.hardresetmedia.com/p/machine-learning-expert-ai-hype-is>.

Publications and Patents

Google Scholar Metrics. **citations: 10,632; h-index: 37**. Patents. **Granted: 4; Under application: 1**.

Full publication list is available at: <https://scholar.google.com/citations?user=PFMB5SsAAAAJ>.

BOOKS, MONOGRAPHS, AND BOOK CHAPTERS

- 2025 Bhaskar Mitra, Henriette Cramer, and Olya Gurevich. *Sociotechnical Implications of Generative Artificial Intelligence for Information Access*. Chapter in Information Access in the Era of Generative AI, Springer.
- 2021 Bhaskar Mitra. *Neural Methods for Effective, Efficient, and Exposure-Aware Information Retrieval*. PhD thesis, University College London, UK. **(PhD dissertation)**
- 2018 Bhaskar Mitra and Nick Craswell. *An Introduction to Neural Information Retrieval*. In Foundations and Trends® in Information Retrieval, Now Publishers, Inc. **(500+ citations)**
- 2014 Amit Agarwal, Eldar Akchurin, Chris Basoglu, Guoguo Chen, Scott Cyphers, Jasha Droppo, Adam Eversole, Brian Guenter, Mark Hillebrand, Xuedong Huang, Zhiheng Huang, Vladimir Ivanov, Alexey Kamenev, Philipp Kranen, Oleksii Kuchaiev, Wolfgang Manousek, Avner May, Bhaskar Mitra, Olivier Nano, Gaizka Navarro, Alexey Orlov, Marko Padmilac, Hari Parthasarathi, Baolin Peng, Alexey Reznichenko, Frank Seide, Michael Seltzer, Malcolm Slaney, Andreas Stolcke, Huaming Wang, Kaisheng Yao, Dong Yu, Yu Zhang, and Geoffrey Zweig. *An introduction to computational networks and the computational network toolkit*. Microsoft Technical Report MSR-TR-2014-112. **(450+ citations)**

CONFERENCE

- Anja Thieme, Rita Faia Marques, Martin Grayson, Sidhika Balachandar, Cameron Tyler Cassidy, Madiha Zahrah Choksi, Camilla Longden, Reeda Shimaz Huda, Nicholas Ilave Kalovwe, Christina Mallon, Courtney Mansperger, Daniela Massiceti, Bhaskar Mitra, Ruth Mueni Nzioka, Ioana Tanase, Yuzhe You, and Cecily Morrison. *Human-Centred AI Pipelines: Designing a Platform for Engaging Communities Meaningfully in AI Data Practices to Improve Disability Representation in Image Generation Models*. In proc. CHI, ACM (to appear).
- 2026 Jaime Arguello, Fernando Diaz, Maik Fröbe, To Eun Kim, and Bhaskar Mitra. *Overview of the TREC 2025 Tip-of-the-Tongue Track*. In proc. TREC.
- 2025 Hossein Rahmani, Varsha Ramineni, Emine Yilmaz, Nick Craswell, and Bhaskar Mitra. *Towards Understanding Bias in Synthetic Data for Evaluation*. In proc. CIKM, ACM.

- 2025 Yifan He, To Eun Kim, Fernando Diaz, Jaime Arguello, and Bhaskar Mitra. *Tip of the Tongue Query Elicitation for Simulated Evaluation*. In proc. SIGIR, ACM.
- 2025 Hossein Rahmani, Emine Yilmaz, Nick Craswell, and Bhaskar Mitra. *JudgeBlender: Ensembling Automatic Relevance Judgments*. In proc. TheWebConf, ACM.
- 2025 Hossein Rahmani, Xi Wang, Emine Yilmaz, Nick Craswell, Bhaskar Mitra, and Paul Thomas. *SynDL: A Large-Scale Synthetic Test Collection for Passage Retrieval*. In proc. TheWebConf, ACM.
- 2025 Jaime Arguello, Samarth Bhargav, Fernando Diaz, To Eun Kim, Yifan He, Evangelos Kanoulas, and Bhaskar Mitra. *Overview of the TREC 2024 Tip-of-the-Tongue Track*. In proc. TREC.
- 2024 Haolun Wu, Ye Yuan, Liana Mikaelyan, Alexander Meulemans, Xue Liu, James Hensman, and Bhaskar Mitra. *Structured Entity Extraction Using Large Language Models*. In proc. EMNLP.
- 2024 Paul Thomas, Seth Spielman, Nick Craswell, and Bhaskar Mitra. *Large language models can accurately predict searcher preferences*. In proc. SIGIR, ACM. **(200+ citations)**
- 2024 Hossein Rahmani, Nick Craswell, Emine Yilmaz, Bhaskar Mitra, and Daniel Campos. *Synthetic Test Collections for Retrieval Evaluation*. In proc. SIGIR, ACM.
- 2024 Haolun Wu, Bhaskar Mitra, and Nick Craswell. *Towards Group-aware Search Success*. In proc. ICTIR, ACM.
- 2024 Anna Gausen, Bhaskar Mitra, and Siân Lindley. *A Framework for Exploring the Consequences of AI-Mediated Enterprise Knowledge Access and Identifying Risks to Workers*. In proc. FAccT, ACM.
- 2024 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Hossein Rahmani, Daniel Campos, Jimmy Lin, Ellen Voorhees, and Ian Soboroff. *Overview of the TREC 2023 Deep Learning Track*. In proc. TREC.
- 2024 Jaime Arguello, Samarth Bhargav, Fernando Diaz, Evangelos Kanoulas, and Bhaskar Mitra. *Overview of the TREC 2023 Tip-of-the-Tongue Track*. In proc. TREC.
- 2023 Amifa Raj, Bhaskar Mitra, Nick Craswell, and Michael Ekstrand. *Patterns of Gender-Specializing Query Reformulation*. In proc. SIGIR, ACM.
- 2023 Chirag Shah, Ryen White, Paul Thomas, Bhaskar Mitra, Shawon Sarkar, and Nicholas Belkin. *Taking Search to Task*. In proc. CHIIR, ACM.
- 2023 Amin Bigdeli, Negar Arabzadeh, Shirin Seyedsalehi, Bhaskar Mitra, Morteza Zihayat, and Ebrahim Bagheri. *De-Biasing Relevance Judgements for Fair Ranking*. In proc. ECIR, Springer.
- 2023 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Jimmy Lin. *Overview of the TREC 2022 Deep Learning Track*. In proc. TREC. **(850+ citations)**
- 2022 Haolun Wu*, Bhaskar Mitra*, Chen Ma, and Xue Liu (*equal contributions). *Joint Multisided Exposure Fairness for Recommendation*. In proc. SIGIR, ACM. **(50+ citations)**
- 2022 Daniel Cohen, Kevin Du, Bhaskar Mitra, Laura Mercurio, Navid Rekabsaz, and Carsten Eickhoff. *Inconsistent Ranking Assumptions in Medical Search and Their Downstream Consequences*. In proc. SIGIR, ACM.
- 2022 Jimmy Lin, Daniel Campos, Nick Craswell, Bhaskar Mitra, and Emine Yilmaz. *Fostering Coopetition While Plugging Leaks: The Design and Implementation of the MS MARCO Leaderboards*. In proc. SIGIR, ACM.
- 2022 Ruohan Li, Jianxiang Li, Bhaskar Mitra, Fernando Diaz, and Asia Biega. *Exposing query identification for Search Transparency*. In proc. TheWebConf, ACM.
- 2022 Gabriella Kazai, Bhaskar Mitra, Anlei Dong, Nick Craswell, and Linjun Yang. *Less is Less: When Are Snippets Insufficient for Human vs Machine Relevance Estimation?* In proc. ECIR, Springer.
- 2022 Nicola Neophytou, Bhaskar Mitra, and Catherine Stinson. *Revisiting Popularity and Demographic Biases in Recommender Evaluation and Effectiveness*. In proc. ECIR, Springer.
- 2022 Shirin SeyedSalehi, Amin Bigdeli, Negar Arabzadeh, Bhaskar Mitra, Morteza Zihayat, and Ebrahim Bagheri. *Bias-aware Fair Neural Ranking for Addressing Stereotypical Gender Biases*. In proc. EDBT.
- 2022 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Jimmy Lin. *Overview of the TREC 2021 Deep Learning Track*. In proc. TREC.
- 2021 Negar Arabzadeh, Bhaskar Mitra, and Ebrahim Bagheri. *MS MARCO Chameleons: Challenging the MS MARCO Leaderboard with Extremely Obstinate Queries*. In proc. CIKM, ACM.
- 2021 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Jimmy Lin. *MS MARCO: Benchmarking ranking models in the large-data regime*. In proc. SIGIR, ACM. **(50+ citations)**
- 2021 Daniel Cohen, Bhaskar Mitra, Oleg Lesota, Navid Rekabsaz, and Carsten Eickhoff. *Not All Relevance Scores are Equal: Efficient Uncertainty and Calibration Modeling for Deep Retrieval Models*. In proc. SIGIR, ACM.
- 2021 Sebastian Hofstätter, Bhaskar Mitra, Hamed Zamani, Nick Craswell, and Allan Hanbury. *Intra-document cascading: Learning to select passages for neural document ranking*. In proc. SIGIR, ACM. **(50+ citations)**

- 2021 Bhaskar Mitra, Sebastian Hofstätter, Hamed Zamani, and Nick Craswell. *Improving Transformer-Kernel Ranking Model Using Conformer and Query Term Independence*. In proc. SIGIR, ACM.
- 2021 Jimmy Lin, Daniel Campos, Nick Craswell, Bhaskar Mitra, and Emine Yilmaz.. *Significant Improvements over the State of the Art? A Case Study of the MS MARCO Document Ranking Leaderboard*. In proc. SIGIR, ACM.
- 2021 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, Ellen Voorhees, and Ian Soboroff. *Trec deep learning track: Reusable test collections in the large data regime*. In proc. SIGIR, ACM. **(50+ citations)**
- 2021 Jaime Arguello, Adam Ferguson, Emery Fine, Bhaskar Mitra, Hamed Zamani, and Fernando Diaz. *Tip of the Tongue Known-Item Retrieval: A Case Study in Movie Identification*. In proc. CHIIR, ACM. **(50+ citations)**
- 2021 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, and Daniel Campos. *Overview of the TREC 2020 Deep Learning Track*. In proc. TREC.
- 2021 Bhaskar Mitra, Sebastian Hofstatter, Hamed Zamani, and Nick Craswell. *Conformer-Kernel with Query Term Independence at TREC 2020 Deep Learning Track*. In proc. TREC.
- 2020 Fernando Diaz, Bhaskar Mitra, Michael D Ekstrand, Asia J Biega, and Ben Carterette. *Evaluating Stochastic Rankings with Expected Exposure*. In proc. CIKM, ACM. **(🏆 Best full paper nominee, 200+ citations)**
- 2020 Nick Craswell, Daniel Campos, Bhaskar Mitra, Emine Yilmaz, and Bodo Billerbeck. *ORCAS: 18 Million Clicked Query-Document Pairs for Analyzing Search*. In proc. CIKM, ACM. **(100+ citations)**
- 2020 Hamed Zamani, Bhaskar Mitra, Everest Chen, Gord Lueck, Fernando Diaz, Paul N Bennett, Nick Craswell, and Susan T Dumais. *Analyzing and Learning from User Interactions for Search Clarification*. In proc. SIGIR, ACM. **(100+ citations)**
- 2020 Sebastian Hofstätter, Hamed Zamani, Bhaskar Mitra, Nick Craswell, and Allan Hanbury. *Local Self-Attention over Long Text for Efficient Document Retrieval*. In proc. SIGIR, ACM. **(100+ citations)**
- 2020 Emine Yilmaz, Nick Craswell, Bhaskar Mitra, and Daniel Campos. *On the Reliability of Test Collections for Evaluating Systems of Different Types*. In proc. SIGIR, ACM.
- 2020 Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Ellen M Voorhees. *Overview of the TREC 2019 Deep Learning Track*. In proc. TREC. **(50+ citations)**
- 2020 Bhaskar Mitra and Nick Craswell. *Duet at TREC 2019 Deep Learning Track*. In proc. TREC.
- 2019 Corby Rosset, Bhaskar Mitra, Chenyan Xiong, Nick Craswell, Xia Song, and Saurabh Tiwary. *An Axiomatic Approach to Regularizing Neural Ranking Models*. In proc. SIGIR, ACM.
- 2018 Hamed Zamani, Bhaskar Mitra, Xia Song, Nick Craswell, and Saurabh Tiwary. *Neural Ranking Models with Multiple Document Fields*. In proc. WSDM, ACM. **(100+ citations)**
- 2018 Daniel Cohen, Bhaskar Mitra, Katja Hofmann, and Bruce Croft. *Cross Domain Regularization for Neural Ranking Models Using Adversarial Learning*. In proc. SIGIR, ACM. **(🏆 Best short paper award, 50+ citations)**
- 2018 Corby Rosset, Damien Jose, Gargi Ghosh, Bhaskar Mitra, and Saurabh Tiwary. *Optimizing Query Evaluations Using Reinforcement Learning for Web Search*. In proc. SIGIR, ACM.
- 2017 Christophe Van Gysel, Bhaskar Mitra, Matteo Venzani, Roy Rosemarin, Grzegorz Kukla, Piotr Grudzien, and Nicola Cancedda. *Reply With: Proactive Recommendation of Email Attachments*. In proc. CIKM, ACM.
- 2017 Bhaskar Mitra Fernando Diaz, and Nick Craswell. *Learning to Match Using Local and Distributed Representations of Text for Web Search*. In proc. WWW. **(600+ citations)**
- 2017 Federico Nanni, Bhaskar Mitra, Matt Magnusson, and Laura Dietz. *Benchmark for Complex Answer Retrieval*. In proc. ICTIR, ACM.
- 2017 Bhaskar Mitra, Fernando Diaz, and Nick Craswell. *Luandri: A Clean Lua Interface to the Indri Search Engine*. In proc. SIGIR, ACM.
- 2016 Fernando Diaz, Bhaskar Mitra, and Nick Craswell. *Query expansion with Locally-Trained Word Embeddings*. In proc. ACL. **(400+ citations)**
- 2016 Eric Nalisnick, Bhaskar Mitra, Nick Craswell, and Rich Caruana. *Improving Document Ranking with Dual Word Embeddings*. In proc. WWW. **(200+ citations)**
- 2015 Bhaskar Mitra. *Exploring Session Context using Distributed Representations of Queries and Reformulations*. In proc. SIGIR, ACM. **(100+ citations)**
- 2015 Bhaskar Mitra and Nick Craswell. *Query Auto-Completion for Rare Prefixes*. In proc. CIKM, ACM. **(100+ citations)**
- 2014 Kajta Hofmann, Bhaskar Mitra, Filip Radlinski, and Milad Shokouhi. *An Eye-Tracking Study of User Interactions with Query Auto Completion*. In proc. CIKM, ACM. **(50+ citations)**
- 2014 Bhaskar Mitra, Milad Shokouhi, Filip Radlinski, and Katja Hofmann. *On User Interactions with Query Auto-Completion*. In proc. SIGIR, ACM. **(50+ citations)**

JOURNAL

- 2025 Bhaskar Mitra. *Emancipatory Information Retrieval*. In IRRJ.
- 2025 Bhaskar Mitra. *Search and Society: Reimagining Information Access for Radical Futures*. In IRRJ.
- 2025 Fernando Diaz, Michael Ekstrand, and Bhaskar Mitra. *Recall, Robustness, and Lexicographic Evaluation*. In TORS, ACM.
- 2024 Haolun Wu, Yansen Zhang, Chen Ma, Fuyuan Lyu, Bowei He, Bhaskar Mitra, and Xue Liu. *Result Diversification in Search and Recommendation: A Survey*. In TKDE, IEEE.
- 2017 Haolun Wu, Chen Ma, Bhaskar Mitra, Fernando Diaz, and Xue Liu. *A Multi-objective Optimization Framework for Multi-stakeholder Fairness-aware Recommendation*. In TOIS, ACM. **(100+ citations)**
- 2017 Nick Craswell, Bruce Croft, Maarten de Rijke, Jiafeng Guo, and Bhaskar Mitra. *Neural information retrieval: introduction to the special issue*. In the special issue of the Information Retrieval Journal on neural information retrieval, Springer Nature.

WORKSHOP

- Hossein A Rahmani, Emine Yilmaz, Nick Craswell, Bhaskar Mitra, Paul Thomas, Charles Clarke, Mohammad
- 2024 Aliannejadi, Clemencia Siro, and Guglielmo Faggioli. *LLMJudge: LLMs for Relevance Judgments*. In proc. LLM4Eval: The First Workshop on Large Language Models for Evaluation in Information Retrieval at SIGIR.
- Andrew Gordon, Carina Negreanu, José Cambronero, Rasika Chakravarthy, Ian Drosos, Hao Fang, Bhaskar Mitra,
- 2024 Hannah Richardson, Advait Sarkar, Stephanie Simmons, Jack Williams, and Ben Zorn. *Co-audit: tools to help humans double-check AI-generated content*. In proc. Workshop on the intersection of HCI and PL at PLATEAU.
- Ida Larsen-Ledet, Bhaskar Mitra, and Siân Lindley. *Ethical and Social Considerations in Automatic Expert Identification and People Recommendation in Organizational Knowledge Management Systems*. In proc. FAccTRec Workshop on Responsible Recommendation at RecSys.
- 2022 Surya Kallumadi, Bhaskar Mitra, and Tereza Iofciu. *A Line in the Sand: Recommendation or Ad-hoc Retrieval?*. In proc. RecSys'18.
- 2018 Navid Rekabsaz, Bhaskar Mitra, Mihai Lupu, and Allan Hanbury. *Toward Incorporation of Relevant Documents in word2vec*. In proc. Neu-ir'17: Neural information retrieval workshop at SIGIR.
- 2017 Bhaskar Mitra, Grady Simon, Jianfeng Gao, Nick Craswell, and Li Deng. *A Proposal for Evaluating Answer Distillation from Web Data*. In proc. Second WebQA Workshop at SIGIR.
- 2016

PREPRINTS

- 2026 Sajad Ebrahimi, Bhaskar Mitra, Negar Arabzadeh, Ye Yuan, Haolun Wu, Fattane Zarrinkalam, and Ebrahim Bagheri. *From Noise to Order: Learning to Rank via Denoising Diffusion*. ArXiv preprint arXiv:2602.11453.
- 2026 Bhaskar Mitra, Nicola Neophytou, and Sireesh Gururaja. *Information Access of the Oppressed: A Problem-Posing Framework for Envisioning Emancipatory Information Access Platforms* 🍌🍌🍌. ArXiv preprint arXiv:2601.09600.
- Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Bhaskar
- 2025 Mitra, Paul Thomas, and Emine Yilmaz. *Judging the Judges: A Collection of LLM-Generated Relevance Judgements*. ArXiv preprint arXiv:2502.13908.
- 2024 Karina Cortiñas Lorenzo, Siân Lindley, Ida Larsen-Ledet, and Bhaskar Mitra. *Through the Looking-Glass: Transparency Implications and Challenges in Enterprise AI Knowledge Systems*. ArXiv preprint arxiv:2401.09410
- 2023 Ouail Kitouni, Niklas Nolte, James Hensman, and Bhaskar Mitra. *DiSK: A Diffusion Model for Structured Knowledge*. ArXiv preprint arXiv:2312.05253.
- 2022 Sebastian Hofstätter, Nick Craswell, Bhaskar Mitra, Hamed Zamani, and Allan Hanbury. *Are We There Yet? A Decision Framework for Replacing Term-Based Retrieval with Dense Retrieval Systems*. ArXiv preprint arXiv:2206.12993.
- 2020 Bhaskar Mitra, Sebastian Hofstatter, Hamed Zamani, and Nick Craswell. *Conformer-Kernel with Query Term Independence for Document Retrieval*. arXiv preprint arXiv:2007.10434.
- 2020 Jason Ingyu Choi, Surya Kallumadi, Bhaskar Mitra, Eugene Agichtein, and Faizan Javed. *Semantic Product Search for Matching Structured Product Catalogs in E-Commerce*. arXiv preprint arXiv:2008.08180.
- Bhaskar Mitra, Corby Rosset, David Hawking, Nick Craswell, Fernando Diaz, and Emine Yilmaz. *Incorporating query term independence assumption for efficient retrieval and ranking using deep neural networks*. arXiv preprint arXiv:1907.03693.
- 2019 Bhaskar Mitra and Nick Craswell. *An updated duet model for passage re-ranking*. arXiv preprint arXiv:1903.07666. **(50+ citations)**
- 2019

- 2018 Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. *MS MARCO: A human generated machine reading comprehension dataset*. arXiv preprint arXiv:1611.09268. **(3,500+ citations)**
- 2017 Bhaskar Mitra and Nick Craswell. *Neural Models for Information Retrieval*. arXiv preprint arXiv:1705.01509. **(250+ citations)**
- 2016 Bhaskar Mitra, Eric Nalisnick, Nick Craswell, and Rich Caruana. *A dual embedding space model for document ranking*. arXiv preprint arXiv:1602.01137. **(150+ citations)**

CONFERENCE AND WORKSHOP REPORTS

- 2025 Bhaskar Mitra. *ACM SIGIR Annual Business Meeting 2025: Secretary's Notes*. In SIGIR Forum, ACM (to appear).
- Matthias Hagen, Josiane Mothe, Smaranda Muresan, Martin Potthast, Min Zhang, Benno Stein, Qinqyao Ai, Mohammad Aliannejadi, Liesbeth Allein, Avishek Anand, Sophia Althammer, Nolwenn Bernard, Arjen de Vries, Niklas Deckers, Gianluca Demartini, Laura Dietz, Carsten Eickhoff, Maik Fröbe, Norbert Fuhr, Marcel Gohsen, Michael Granitzer, Faegheh Hasibi, Sebastian Heineking, Djoerd Hiemstra, Adam Jatowt, Abhinav Joshi, Johannes Kiesel, Wojciech Kusa, Sean MacAvaney, Bhaskar Mitra, Jian-Yun Nie, Heather O'Brien, Birte Platow, Mark Sanderson, Harrison Scells, Damiano Spina, Benno Stein, Johanne Trippas, Stefan Voigt, and Guido Zuccon. *Retrieval-Augmented Generation – The Future of Search? (Dagstuhl Perspectives Workshop 25391)*. In Dagstuhl Reports (to appear).
- 2025 Clemencia Siro, Hossein Rahmani, Mohammad Aliannejadi, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Bhaskar Mitra, Paul Thomas, and Emine Yilmaz. *LLM4Eval: Large Language Model for Evaluation in IR*. In proc. SIGIR, ACM.
- 2025 Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charles L.A. Clarke, Guglielmo Faggioli, Bhaskar Mitra, Paul Thomas, and Emine Yilmaz. *LLM4Eval @ WSDM 2025: Large Language Model for Evaluation in Information Retrieval*. In proc. WSDM, ACM.
- Johanne Trippas, Shane Culpepper, Mohammad Aliannejadi, James Allan, Enrique Amigó, Jaime Arguello, Leif Azzopardi, Peter Bailey, Jamie Callan, Rob Capra, Nick Craswell, Bruce Croft, Jeff Dalton, Gianluca Demartini, Laura Dietz, Zhicheng Dou, Carsten Eickhoff, Michael Ekstrand, Nicola Ferro, Norbert Fuhr, Dorota Glowacka, Faegheh Hasibi, Danula Hettiachchi, Rosie Jones, Jaap Kamps, Noriko Kando, Sarvnaz Karimi, Makoto P Kato, Bevan Koopman, Yiqun Liu, Chenglong Ma, Joel Mackenzie, Maria Maistro, Jiaxin Mao, Dana McKay, Bhaskar Mitra, Stefano Mizzaro, Alistair Moffat, Josiane Mothe, Iadh Ounis, Lida Rashidi, Yongli Ren, Mark Sanderson, Rodrygo Santos, Falk Scholer, Chirag Shah, Laurianne Sitbon, Ian Soboroff, Damiano Spina, Paul Thomas, Juli ´an Urbano, Arjen de Vries, Ryen White, Abby Yuan, Hamed Zamani, Oleg Zendel, Min Zhang, Justin Zobel, Shengyao Zhuang, and Guido Zuccon. *Report from the Fourth Strategic Workshop on Information Retrieval in Lorne (SWIRL 2025)*. In SIGIR Forum, ACM.
- 2024 Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Bhaskar Mitra, Paul Thomas, and Emine Yilmaz. *Report on the 1st Workshop on Large Language Model for Evaluation in Information Retrieval (LLM4Eval 2024) at SIGIR 2024*. In SIGIR Forum, ACM.
- 2024 Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Bhaskar Mitra, Paul Thomas, and Emine Yilmaz. *LLM4Eval: Large Language Model for Evaluation in IR*. In proc. SIGIR, ACM.
- 2024 Maik Fröbe, Joel Mackenzie, Bhaskar Mitra, Franco Maria Nardini, and Martin Potthast. *ReNeuIR at SIGIR 2024: The third workshop on reaching efficiency in neural information retrieval*. In proc. SIGIR, ACM.
- 2024 Leif Azzopardi, Charles Clarke, Paul Kantor, Bhaskar Mitra, Johanne Trippas, Zhaochun Ren, and others. *Report on The Search Futures Workshop at ECIR 2024*. In SIGIR Forum, ACM.
- 2024 Leif Azzopardi, Charles Clarke, Paul Kantor, Bhaskar Mitra, Johanne Trippas, and Zhaochun Ren. *The search futures workshop*. In proc. ECIR, Springer.
- Chirag Shah, Torsten Suel, Fernando Diaz, Bhaskar Mitra, Bárbara Poblete, Hussein Suleman, and Suzan Verberne. *Report on the 44th international ACM SIGIR conference on research and development in information retrieval (SIGIR 2021)*. In SIGIR Forum, ACM.
- 2020 Laura Dietz, Bhaskar Mitra, Jeremy Pickens, and others. *Report on the First HIPstIR Workshop on the Future of Information Retrieval*. In SIGIR Forum, ACM.
- 2017 Nick Craswell, Bruce Croft, Maarten de Rijke, Jiafeng Guo, and Bhaskar Mitra. *Report on the Second SIGIR Workshop on Neural Information Retrieval (Neu-IR'17)*. In SIGIR Forum, ACM.
- 2016 Nick Craswell, Bruce Croft, Jiafeng Guo, Bhaskar Mitra, and Maarten de Rijke. *Report on the SIGIR 2016 Workshop on Neural Information Retrieval (Neu-IR)*. In SIGIR Forum, ACM.

TUTORIALS

- 2018 Tom Kenter, Alexey Borisov, Christophe Van Gysel, Mostafa Dehghani, Maarten de Rijke, and Bhaskar Mitra. *Neural networks for information retrieval*. In proc. ECIR, Springer.
- 2018 Tom Kenter, Alexey Borisov, Mostafa Dehghani, Maarten de Rijke, and Bhaskar Mitra. *Neural networks for information retrieval*. In proc. WSDM, ACM.
- 2017 Tom Kenter, Alexey Borisov, Christophe Van Gysel, Mostafa Dehghani, Maarten de Rijke, and Bhaskar Mitra. *Neural networks for information retrieval*. In proc. SIGIR, ACM. **(50+ citations)**
- 2017 Bhaskar Mitra and Nick Craswell. *Neural text embeddings for information retrieval*. In proc. WSDM, ACM. **(50+ citations)**

CONFERENCE AND WORKSHOP PROCEEDINGS

- 2025 Keping Bi, Qingshan Li, Evangelos Kanoulas, Yadan Luo, and Bhaskar Mitra. *SIGIR-AP 2025: Proceedings of the 2025 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*. ACM.
- 2024 Clemencia Siro, Mohammad Aliannejadi, Hossein Rahmani, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Bhaskar Mitra, Paul Thomas, and Emine Yilmaz. *Proceedings of The First Workshop on Large Language Models for Evaluation in Information Retrieval (LLM4Eval 2024)*. CEUR Workshop Proceedings (CEUR-WS.org).
- 2023 Debasis Ganguly, Srijoni Majumdar, Bhaskar Mitra, Parth Gupta, Surupendu Gangopadhyay, and Prasenjit Majumder. *Proceedings of the 15th Annual Meeting of the Forum for Information Retrieval Evaluation*. ACM.

PATENTS

- 2023 Bhaskar Mitra, Yordan Zaykov, John Winn, and James Hensman. *Incorporating Structured Knowledge in Neural Networks*. US Patent application.
- 2023 Corby Rosset, Bhaskar Mitra, David Hawking, Nick Craswell, Fernando Diaz, and Emine Yilmaz. *Neural network for search retrieval and ranking*. US Patent 11,615,149.
- 2023 Amy Huyen Phuoc Nguyen, Bhaskar Mitra, Christophe Jacky Henri Van Gysel, Grzegorz Stanislaw Kukla, Lynn Carter Ayres, Mark Rolland Knight, Matteo Venzani, Nicola Cancedda, Rachel Elizabeth Sirkin, Robin Michael Thomas, Roy Rosemarin, Shobana Balakrishnan, Sri Ramya Mallipudi, Tariq Sharif, and Yamin Wang. *Intelligent query system for attachments*. US Patent 11,556,548.
- 2016 Bhaskar Mitra, Elbio Renato Abib, Fabio Eigi Imada, and Yu Jiao. *Inferring entity attribute values*. US Patent 9,501,503.
- 2014 Rupesh Rasiklal Mehta, Sree Hari Nagaralu, Anjana Das, and Bhaskar Mitra. *Utilization of features extracted from structured documents to improve search relevance*. US Patent 8,788,436.

MEDIA ARTICLES

- 2026 Bhaskar Mitra. *Why I'm Leaving Big Tech*. In Disjunctions magazine.

Professional Activities

COMMUNITY LEADERSHIP

- ACM SIGIR Secretary, 2025-2028.
- Associate Editor, ACM Transactions on Information Systems, 2024-2027.
- ACM SIGIR Community Relations Coordinator, 2022-2025.
- Guest editor: Special Issue of the Information Retrieval Journal on Neural Information Retrieval, with Nick Craswell, Bruce Croft, Maarten de Rijke, and Jiafeng Guo, 2017.

CONFERENCE CHAIRING

- Responsible Web Track Chair, WebConf, 2026.
- IR for Good Chair, ECIR, 2026.
- Program Chair, SIGIR-AP, 2025.
- Best paper Chair, SIGIR-AP, 2024.
- Resource & Reproducibility Chair, SIGIR, 2024.
- Industry Track Chair, FIRE, 2023.
- Tutorials Chair, ECIR, 2023.
- PhD Symposium Chair, CIKM, 2022.
- Diversity, Equity, and Inclusion Sponsorship Chair, CIKM, 2021.
- Diversity, Equity, and Inclusion Chair, SIGIR, 2021.
- Virtual chair, SIGIR, 2021.
- Session chair: IR Models, SIGIR, 2021.
- Session Chair: Deep learning in practice, Industry track at WWW, 2016.

- Best Paper Committee Member: ECIR (2025), SIGIR (2022).
- Senior Area Chair, Senior Program Committee Member, and Program Committee Member: ACL, CHIIR, CIKM, EMNLP, FAccT, ICTIR, NAACL, SIGIR, TREC, WebConf, WSDM, and other conferences.

SHARED TASKS

- Coordinator: Tip-of-the-Tongue task at the NII Test Collection for IR Systems (NTCIR), (2026).
- Coordinator: Tip-of-the-Tongue track at the Text Retrieval Conference (TREC), 2023-2025.
- Coordinator: Deep Learning Track at the Text Retrieval Conference (TREC), 2019 - 2023.
- Organizer: Microsoft MACHine Reading COMprehension (MS MARCO) passage and document ranking challenge, with Payal Bajaj, Daniel Campos, Nick Craswell, and others, 2018.

INVITED STRATEGIC WORKSHOP PARTICIPATION

- Dagstuhl Seminar: Retrieval-Augmented Generation - The Future of Search?, 2025.
- The 4th Strategic Workshop on Information Retrieval in Lorne (SWIRL), 2025.
- Future of Information Retrieval Research in the Age of Generative AI, Computing Community Consortium, 2024.

KEYNOTE ADDRESSES

- Emancipatory Information Retrieval: Radically Reorienting Information Retrieval Research to Resist Corporate and Authoritarian Capture of our Information Ecosystems, at Forum for Information Retrieval Evaluation (FIRE), Virtual, 2025.
- Bias and Beyond: On Generative AI and the Future of Search and Society, at The International Workshop on Algorithmic Bias in Search and Recommendation (BIAS), at ACM SIGIR, Washington DC, USA, July 2024.
- Efficient Machine Learning and Machine Learning for Efficiency in Information Retrieval, TREC, and Beyond, at The Workshop on Reaching Efficiency in Neural Information Retrieval (ReNeuIR), at ACM SIGIR, Madrid, Spain, July 2022.
- Benchmarking for Neural Information Retrieval: MS MARCO, TREC, and Beyond, at The 6th Natural Language Interfaces for the Web of Data (NLIWOD) Workshop, at The 19th International Semantic Web Conference, Virtual, November 2020.
- 5 Lessons Learned from Designing Neural Models for Information Retrieval, at The 10th Recherche d'Information SEmantique (RISE) workshop, at The CORIA-TALN-RJC conference, Rennes, France, May 2018.

INVITED TALKS

- Information Access of the Oppressed: Envisioning Emancipatory Information Access Platforms, at Starling Centre for Just Technologies and Just Societies, Virtual, 2026.
- Information Access of the Oppressed: Envisioning Emancipatory Information Access Platforms, at University College London Department of Information Studies, Virtual, 2026.
- Sociotechnical Implications of RAG for Information Access, at Dagstuhl Seminar 25391: Retrieval-Augmented Generation – The Future of Search?, Wadern, Germany, 2025.
- Emancipatory Information Retrieval, at People and Technology Lab (PATLab), University College Cork, Virtual, 2025.
- Sociotechnical Implications of Generative AI for Information Access, at (Re)defining Responsible AI workshop, MILA, Montréal, Canada, 2024.
- Search and Society: Reimagining Information Access for Radical Futures, at Canadian AI, Guelph, Canada, 2024.
- Joint Multisided Exposure Fairness for Search and Recommendation, at SEA: Search Engines Amsterdam, Virtual, 2023.
- What's next for deep learning for Search?, at Etsy, Virtual, 2022.
- So, You Want to Release a Dataset? Reflections on Benchmark Development, Community Building, and Making Robust Scientific Progress, at Spotify, Virtual, 2022.
- Multisided Exposure fairness for Search and Recommendation, at Lowe's, Virtual, 2022.
- Deep Learning for Effective, Exposure-Aware, and Efficient Information Retrieval, at Microsoft Research, Cambridge, UK, 2022.
- Multisided Exposure fairness for Search and Recommendation, at Microsoft Research, Montreal, Canada, 2022.
- Neural Information Retrieval: In search of meaningful progress, at CIIR Talk Series, University of Massachusetts Amherst, Virtual, 2021.
- Neural Information Retrieval: In search of meaningful progress, at CLIP Colloquium, University of Maryland, Virtual, 2021.
- Neural Models for Document Ranking, at Facebook, Seattle, USA, 2017.
- Neural Models for Document Ranking, at Microsoft Research, Redmond, USA, 2017.
- Neural Models for Document Ranking, at University of Glasgow, Glasgow, UK, 2017.
- Neural Models for Document Ranking, at The 4th International Alexandria Workshop, Hannover, Germany, 2017.
- Neural Models for Information Retrieval, at The NLIP seminar series, University of Cambridge, Cambridge, UK, 2017.
- Using Text Embeddings for Information Retrieval, at University of Glasgow, Glasgow, UK, 2016.
- A Simple Introduction to Word Embeddings, at North Eastern University, Seattle, USA, 2016.
- Vectorland: Brief Notes from Using Text Embeddings for Search, at Search Solutions, London, UK, 2015.

GUEST LECTURES

- Information Retrieval & Society, at University of Amsterdam Information Retrieval Lab, Virtual, 2026.
- Neural Learning to Rank, at Dayananda Sagar College of Engineering, Virtual, 2022.
- Deep Neural Methods for Retrieval, at University College London, Virtual, 2021.
- Neural Learning to Rank, at University College London, Virtual, 2021.
- Deep Neural Methods for Retrieval, at Emory University, Virtual, 2020.
- Neural Learning to Rank, at Emory University, Virtual, 2020.
- A Simple Introduction to Neural Information Retrieval, at University College London, London, UK, 2018.

INVITED PANELIST

- *Global South in AI Governance*, Next Gen AI Forum: Preparing for Tomorrow's AI Governance Today (a pre-summit convening for the India AI Impact Summit), Encode Canada, 2026.
- *AI Ethics and the Tech Industry*, McGill Computer Science Undergraduate Society (CSUS) and Encode Canada, 2025.
- *To Use Generative AI or Not To Use It?*, Women in AI Ethics Plus (WAIE+) AI Debate, 2025.
- *Diversity, Equity, and Inclusion (DEI) lunch*, SIGIR, 2024.
- *From Research To Production*, ECIR Industry Day, 2018.

WORKSHOP ORGANIZATION

- *Justice, Emancipation, Democracy, and Information Access (JEDI): The SIGIR Workshop on Resisting Corporate and Authoritarian Capture of Information Access Platforms*, with Dana Mckay, Michael Ekstrand, Sanne Vrijenhoek, and Maria Murray, SIGIR, 2026.
- *VulGen: International Workshop on Vulnerabilities in Generative Systems for Information Retrieval*, with Shuoqi Sun, Sara Allawati, Laura Dietz, Madhurima Khirbat, Maarten de Rijke, and Damiano Spina, SIGIR, 2026.
- *LLM4Eval: Large Language Model for Evaluation in IR*, with Clemencia Siro, Hossein Rahmani, Mohammad Aliannejadi, Nick Craswell, Charles Clarke, Guglielmo Faggioli, Paul Thomas, and Emine Yilmaz, SIGIR, 2025.
- *LLM4Eval: Large Language Model for Evaluation in IR*, with Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charlie Clarke, Guglielmo Faggioli, Paul Thomas, and Emine Yilmaz, WSDM, 2025.
- *LLM4Eval: Large Language Model for Evaluation in IR*, with Hossein Rahmani, Clemencia Siro, Mohammad Aliannejadi, Nick Craswell, Charlie Clarke, Guglielmo Faggioli, Paul Thomas, and Emine Yilmaz, SIGIR, 2024.
- *ReNeuIR: The Third Workshop on Reaching Efficiency in Neural Information Retrieval*, with Maik Fröbe, Joel Mackenzie, Franco Maria Nardini, and Martin Potthast, SIGIR, 2024.
- *The Search Futures Workshop*, with Leif Azzopardi, Charlie Clarke, Paul Kantor, Johanne Trippas, and Zhaochun Ren, ECIR, 2024.
- *The HIPstIR Workshop: The hip "stick, sand, and paper" retreat on the future of IR*, with Laura Dietz and Jeremy Pickens, 2019.
- *The ACM SIGIR Workshop on Neural Information Retrieval (Neu-IR)*, with Nick Craswell, Bruce Croft, Jiafeng Guo, and Maarten de Rijke, SIGIR, 2017.
- *The ACM SIGIR Workshop on Neural Information Retrieval (Neu-IR)*, with Nick Craswell, Bruce Croft, Jiafeng Guo, and Maarten de Rijke, SIGIR, 2016.

TUTORIALS

- *Learning to Rank for Information Retrieval with Neural Networks*, with Nick Craswell, Emine Yilmaz, and Daniel Campos, at ACM SIGIR/SIGKDD Africa Summer School on Machine Learning for Data Mining and Search (AFIRM), 2020.
- *Deep Learning for Search*, at Forum for Information Retrieval Evaluation (FIRE), 2019.
- *Neural Learning to Rank*, at IVADO Recommender Systems Summer School, 2019.
- *Deep Learning for Information Retrieval*, with Nick Craswell, Emine Yilmaz, and Daniel Campos, at ACM SIGIR/SIGKDD Africa Summer School on Machine Learning for Data Mining and Search (AFIRM), 2019.
- *Neural Networks for Information Retrieval*, with Tom Kenter, Alexey Borisov, Christophe Van Gysel, Mostafa Dehghani, and Maarten de Rijke, ECIR, 2018.
- *Neural Networks for Information Retrieval*, with Tom Kenter, Alexey Borisov, Christophe Van Gysel, Mostafa Dehghani, and Maarten de Rijke, WSDM, 2018.
- *NN4IR: The SIGIR 2017 tutorial on Neural Networks for Information Retrieval*, with Tom Kenter, Alexey Borisov, Christophe Van Gysel, Mostafa Dehghani, and Maarten de Rijke, SIGIR, 2017.
- *The WSDM 2017 Tutorial on Neural Text Embeddings for Information Retrieval*, with Nick Craswell, WSDM, 2017.

Diversity, Equity, and Inclusion

- Serving on the Women in AI Ethics Plus (WAIE+) advisory council (2025-2026).
- Invited panelist at the Diversity, Equity, and Inclusion (DEI) lunch, SIGIR, 2024.
- Served as Diversity, Equity, and Inclusion (DEI) Sponsorship Chair, CIKM, 2021.
- Served as Diversity, Equity, and Inclusion (DEI) Chair, SIGIR, 2021.
- Co-founded grassroot Diversity, Inclusion, and Belonging (DIB) committees at Microsoft that organized regular discussions on DIB related topics among other initiatives.
- Organized tutorials aimed at growing the IR research community in Africa (AFIRM) and India (FIRE).

Travel and volunteering

- Visited Svalbard, one of the world's northernmost inhabited areas that hosts the global seed vault, 2024.
- I have travelled to all seven continents, incl. Antarctica, 2023.
- Race crew, Leg 1 of the Clipper Round The World yacht race (sailed for 34 days from Liverpool, UK to Punte del Este, Uruguay), 2017.
- Open Water Certified diver with multiple dives around the world—incl. at Silfra, Iceland in the rift between the North American and Eurasian tectonic plates, 2016.
- Climbed Mt. Kilimanjaro (Uhuru Peak), highest peak in Africa, 2014.
- Multiple trips as a general volunteer on medical brigades to Tena, Ecuador with Timmy Global Health (2012-14, 2016-17).
- Volunteered on lion rehabilitation projects in Antelope Park, Zimbabwe with African Impact, 2014.

References

- *Mary L. Gray*, Senior Principal Researcher, Microsoft Research, USA (Email: mlg@microsoft.com)
- *Abeba Birhane*, Founder & Principal Investigator, AI Accountability Lab, Trinity College Dublin, Ireland (Email: birhanea@tcd.ie)
- *Fernando Diaz*, Associate Professor, Language Technologies Institute, Carnegie Mellon University, USA (Email: diazf@acm.org)
- *Nick Craswell*, Principal Architect, Microsoft, USA (Email: nickcr@microsoft.com)
- *Emine Yilmaz*, Professor, University College London, UK (Email: emine.yilmaz@ucl.ac.uk)